

Learn over Past, Evolve for Future: Search-based Time-aware Recommendation with Sequential Behavior Data

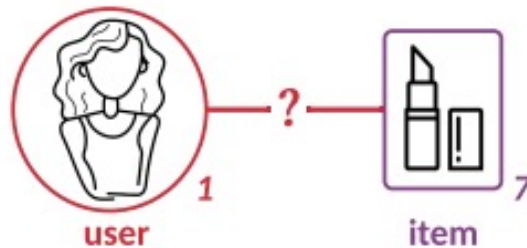
**Jiarui Jin*, Xianyu Chen*, Weinan Zhang, Junjie
Huang, Ziming Feng, Yong Yu**

**Apex Data & Knowledge Management Lab
Shanghai Jiao Tong University**



Personalized Recommendation

- Personalized recommendation is an important part of modern e-commerce.
- Users' needs depend not only on their profile, but also on their recent browsing behavior and regular purchases made in the past.



Existing Methods

- Classical recommendation methods
 - Primary focus on general user interest modeling, ignoring the time factor.
 - Collaborative filtering based models; Factorization machine based models. E.g. Wide & deep learning for recommender systems.
- Sequential pattern-based model
 - Due to hardware and time constraints, it can only be applied to hundreds of lengths of user data.
 - Memory networks; Recurrent neural networks; Temporal point processes. E.g. Recurrent recommender networks.
- Search-Based Model
 - Focus on searching useful information from historical data, ignoring the user's sequential pattern also.
 - E.g. User behavior retrieval for click-through rate prediction.



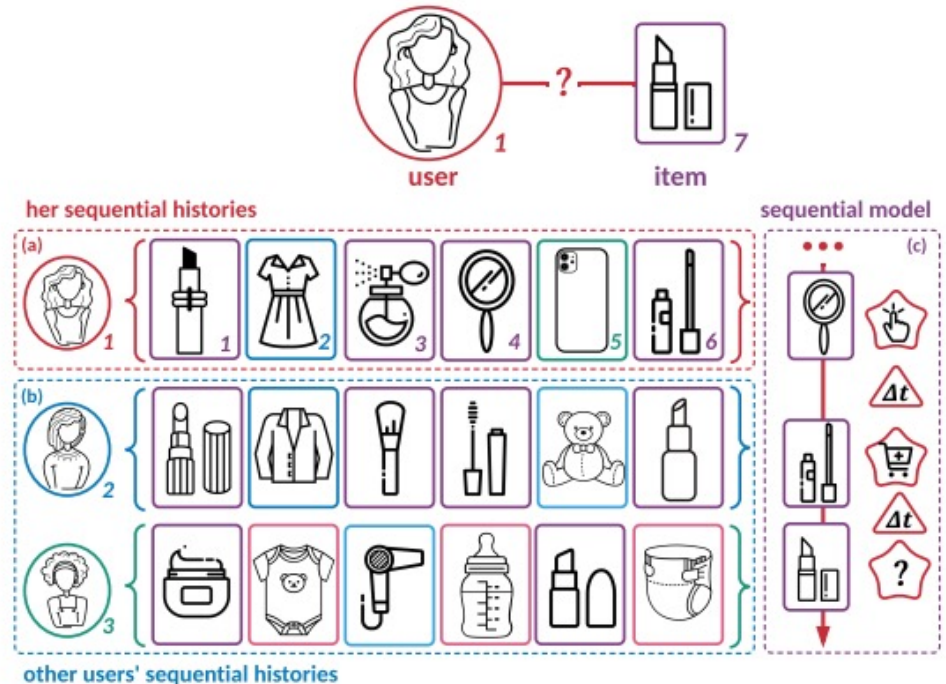
Challenges

- How to incorporate the user's sequential patterns into these search-based models?
- How can labeling information from historical data (i.e., user feedback) be used more effectively in recommendation models?
- How to train both search-based models and predictive models in an end-to-end manner?

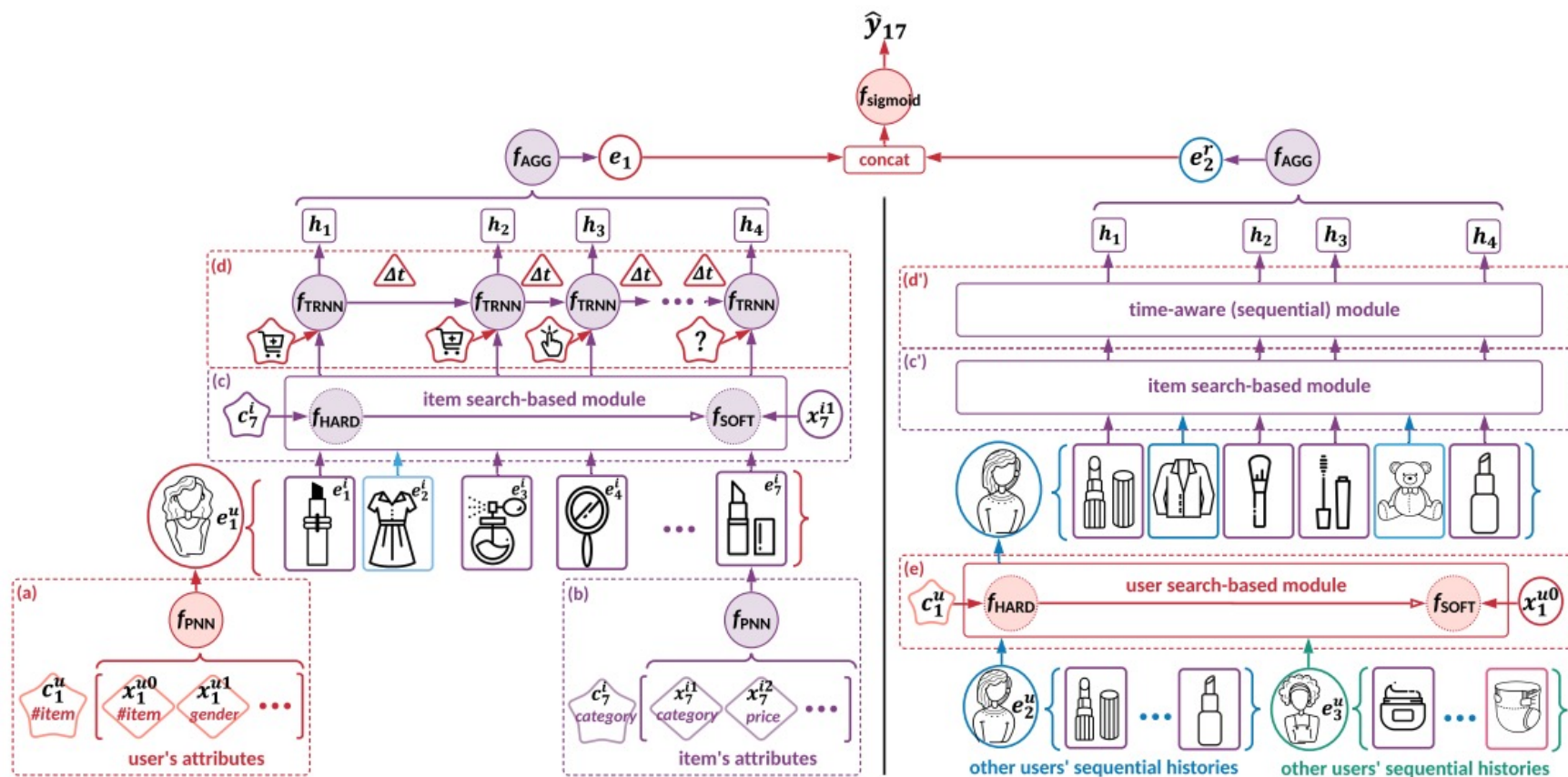


Motivation

- Searching for relevant items is sufficient.
- Users have diverse and drifting interests.
- The user's current behavior may be significantly influenced by previous behavior.



STARec Framework



Object Modeling

- Classical neural networks (e.g., multilayer perceptron (MLP)) have difficulty modeling the rich correlation between features, which can be captured by product-based models such as PNN.

$$e_n^i := f_{\text{PNN}}^i(x_n^i) = v_n^i \odot x_n^i + \sum_{p=1}^P \sum_{p'=p+1}^P (v_p^i \odot v_{p'}^i) x_n^{ip} x_n^{ip'},$$



Search-based Module

- **Item Search:** $\hat{\mathcal{H}}_m := \{i_{n'} | i_{n'} \in \mathcal{H}_m \wedge f_{\text{HARD}}^i(c_{n'}, c_n^i) \geq \epsilon\} \cup \mathcal{H}_m^{\text{RECENT}}$
 - Hard-Search: Based on the number of common categories between items
 - Soft-Search: Based on cosine similarity
 - Adaptive Search:

$$f_{\text{ADA}}^i(c_{n'}, c_n^i, x_{n'}^{i1}, x_n^{i1}) := -\frac{\text{sgn}(|c_{n'}^i - c_n^i|)}{1 - \tau} + f_{\text{softmax}}(\cos(x_{n'}^{i1}, x_n^{i1}), \tau)$$

$$= -\frac{\text{sgn}(|c_{n'}^i - c_n^i|)}{1 - \tau} + \frac{\exp(\cos(x_{n'}^{i1}, x_n^{i1})/\tau)}{\sum_{i_{n''} \in \mathcal{H}_m} \exp(\cos(x_{n''}^{i1}, x_n^{i1})/\tau)},$$

- **User Search:** $\hat{\mathcal{S}}_m := \{u_{m'} | u_{m'} \in \mathcal{U} \wedge f_{\text{HARD}}^u(c_m^u, c_{m'}^u) \geq \eta\},$

$$f_{\text{ADA}}^u(c_{m'}, c_m^u, e_{m'}^u, e_m^u) := -\frac{\text{sgn}(|c_{m'}^u - c_m^u|)}{1 - \iota} + f_{\text{softmax}}(\cos(e_{m'}^u, e_m^u), \iota)$$

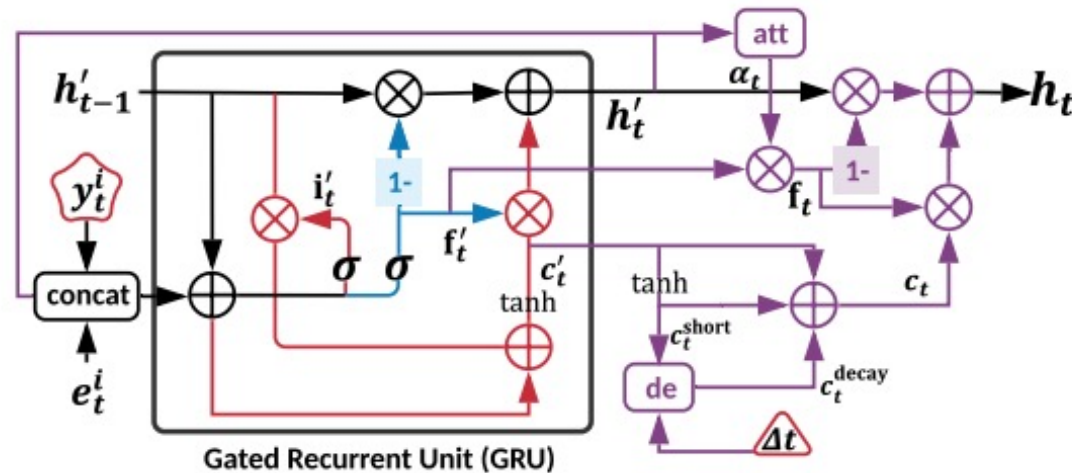
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- For each similar user obtained, we also use the same search-based module to reconstruct their browsing



Time-aware Module

- Input: Concat of vector embedding and previous moment label: $[e_t^i, y_t^i]$
- Cell: Extension of GRU to introduce a time factor



- Aggregation of sequence states

$$e_m = f_{\text{AGG}}(\{h_t | i_t \in \hat{\mathcal{H}}_m\}) = \sigma(W_m \cdot (\sum_{i_t \in \hat{\mathcal{H}}_m} \alpha_t(h_t W_t)) + b_m),$$



Optimization Objective

- Aggregate embeddings for all users (including itself and similar users)

$$\hat{y}_{mn} = \text{sigmoid}(f_{\text{MLP}}([e_m, \{e_{m'}^r | u_{m'} \in \hat{S}_m\}])).$$

- Using log loss to optimize parameters

$$\mathcal{L}_\theta = - \sum_{(u_m, i_n) \in \mathcal{D}} (y_{mn} \cdot \log(\hat{y}_{mn}) + (1 - y_{mn}) \cdot \log(1 - \hat{y}_{mn})),$$



Experiments

- Datasets
 - Tmall, Taobao, Alipay
- Compared Methods
 - Sequential models: LSTM, RRN, STAMP, Time-LSTM, NHP, DUPN, NARM, DIN, DIEN
 - Tower architecture based models: ESMM, ESM², MMoE
 - Search-based model: Sim
- Our Methods
 - STARec, STARec_{time}⁻, STARec_{recent}⁻, STARec_{label}⁺
- Evaluation metric: AUC, ACC, LogLoss



Experiments

- Overall Performance

Recommender	Tmall			Alipay			Taobao		
	AUC	ACC	LogLoss	AUC	ACC	LogLoss	AUC	ACC	LogLoss
LSTM	0.6973	0.7054	0.5854	0.8357	0.7697	0.4713	0.5912	0.4516	0.4411
LSTM⁺_{label}	0.7662	0.7324	0.5291	0.9052	0.8449	0.3738	0.6450	0.5780	0.4094
RRN	0.6973	0.7073	0.5866	0.8429	0.7145	0.4636	0.5102	0.1502	0.4398
Time-LSTM	0.6962	0.6796	0.5865	0.8439	0.8057	0.4874	0.5945	0.4393	0.4397
NHP	0.6979	0.6969	0.5849	0.8490	0.7922	0.4743	0.6003	0.3205	0.4393
DUPN	0.5551	0.6796	0.6269	0.8021	0.7552	0.5243	0.5525	0.1921	0.4471
NARM	0.6396	0.6839	0.6025	0.8422	0.7695	0.4860	0.5972	0.3426	0.4494
STAMP	0.6753	0.6946	0.5829	0.8178	0.7491	0.5137	0.6012	0.3592	0.4448
ESMM	0.5189	0.6796	0.6312	0.7131	0.6856	0.6080	0.5487	0.1774	0.4573
ESM ²	0.5149	0.6796	0.6310	0.7241	0.6930	0.5996	0.5030	0.1520	0.4594
MMoE	0.5060	0.6796	0.6313	0.7119	0.6834	0.6085	0.5501	0.1719	0.4565
DIN	0.6878	0.6946	0.5915	0.8496	0.7692	0.4717	0.5978	0.4422	0.4388
DIEN	0.6892	0.6962	0.5833	0.8474	0.7949	0.4668	0.5963	0.4025	0.4412
SIM	0.7005	0.7094	0.5698	0.8549	0.8069	0.4623	0.6045	0.4538	0.4271
STARec _{time} ⁻	0.6999	0.7097	0.5684	0.8527	0.8046	0.4547	0.6035	0.4525	0.4312
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STARec	0.7204*	0.7150*	0.5471*	0.8624*	0.8142*	0.4410*	0.6126*	0.4629*	0.4211*
STARec_{label}⁺	0.7986*	0.7502*	0.5059*	0.9201*	0.8661*	0.3423*	0.6771*	0.6039*	0.3862*



Experiments

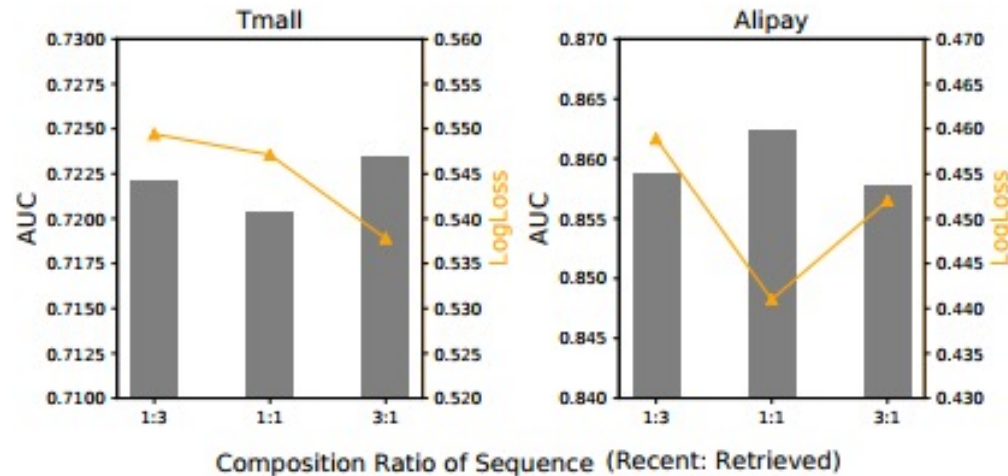
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Experiments

- Composition Ratio of Sequence



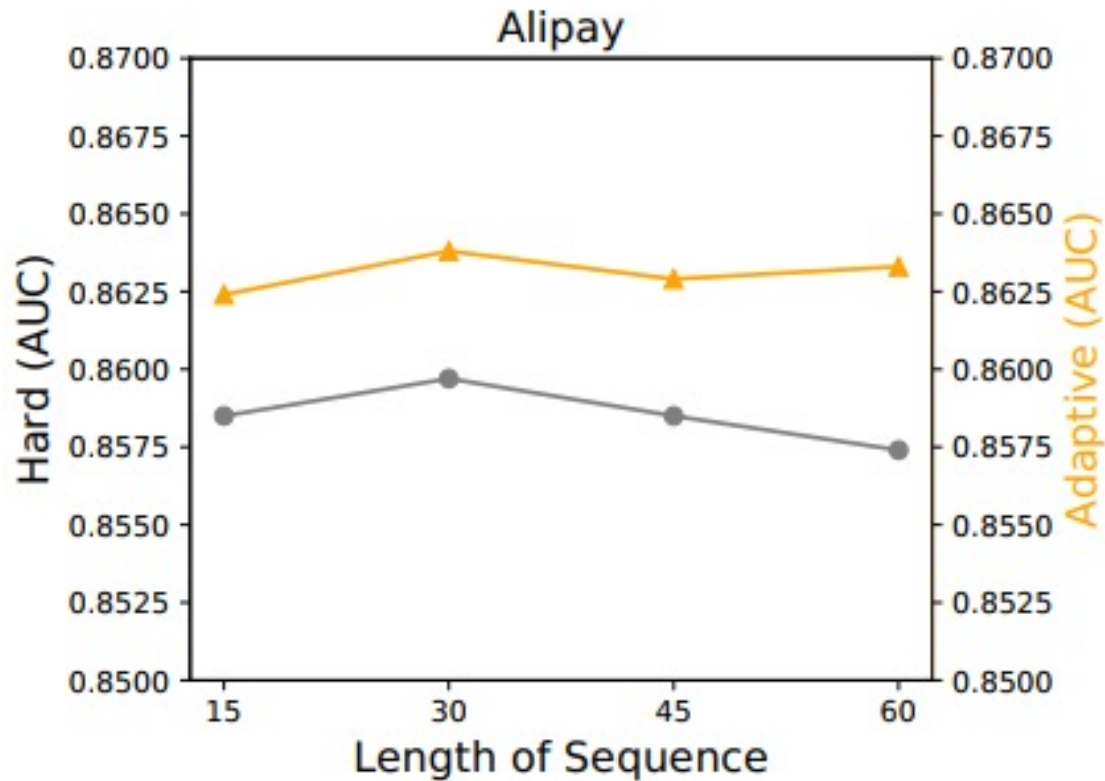
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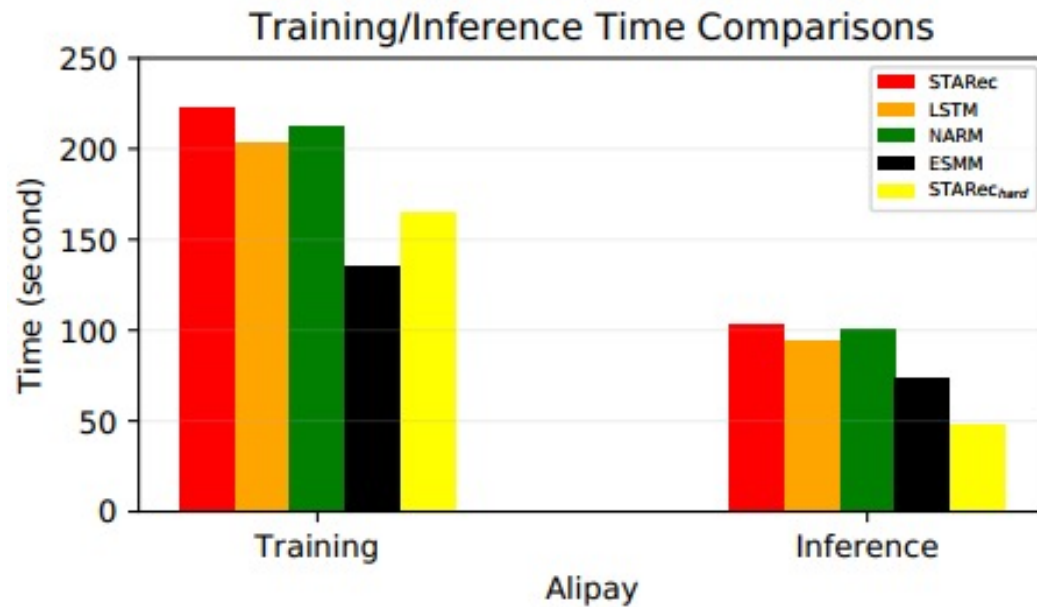
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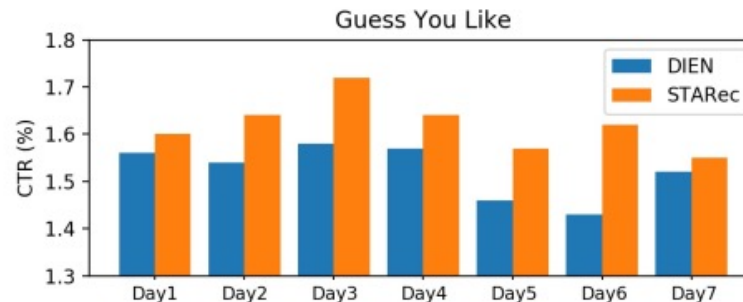


Real-World Deployment

- Offline Evaluation

Recommender	Guess You Like		Information Flow	
	AUC	CTR	AUC	CTR
DIN	0.7828	1.47%	0.8409	1.26%
DIEN	0.8068	1.52%	0.8493	1.30%
STARec	0.8909	1.61%	0.9159	1.32%

- Online Evaluation



Conclusion

- We propose a novel framework named STARec, which captures the user's time-sensitive evolving demands via combining a search-based module and a time-aware module.
- We propose to involve the user's previous feedbacks as input and reveal that this label information can improve the performance.
- We design a new adaptive search mechanism, which gradually transfers from the hard-search strategy to the soft one.



Thanks

Apex Data & Knowledge Management Lab
Shanghai Jiao Tong University

